

Helix Constitution — Universal QWEN.md

Field	Value
Revision	12
Created	2026-05-20
Last modified	2026-06-03T00:00:00Z
Status	active
Status summary	Added §11.4.108 Four-layer fix-verification + runtime-signature-as-definition-of-done mandate mirror (systematic-debugging Phase 4.5, 2026-06-03) — a fix crosses FOUR distinct layers (SOURCE → ARTIFACT → RUNTIME-ON-CLEAN-TARGET → USER-VISIBLE) and “fixed” at one does NOT imply fixed at the next; source-green is the cheapest least-conclusive signal. Every fix declares ONE machine-checkable runtime signature verified on a clean target — this per-fix runtime-signature registry is the SINGLE SOURCE OF TRUTH for “fixed”, REPLACING grep-the-source as the definition of done; gates span all four layers (artifact gate asserts BYTES landed); eliminate the stale-deployment/shadow layer by construction (clean deployment OR prove running-artifact == built-artifact before validation); meta-rule ≥ 3 fixed-but-not-working discoveries = ONE architectural VERIFICATION flaw → fix the pipeline + re-certify on a clean target; a batch is validated only after comprehensive per-item runtime-signature verification on a clean baseline. Composes §11.4.1/.2/.4/.5/.6/.27/.40/.46/.50/.52/.69/.102. §11.4.78–§11.4.107 mirrors continue from earlier Revisions.
Issues	none
Issues summary	—
Fixed	§11.4.108 mirror
Fixed summary	§11.4.107 lands in lockstep with the Constitution.md §11.4.107 addition.
Continuation	—

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How inheritance works

This QWEN.md is for the **Qwen Code CLI agent** (qwen-code). It documents the universal Helix Constitution from the Qwen agent's perspective — the same content seen by Claude Code via CLAUDE.md and by OpenCode / Cursor / generic AI tooling via AGENTS.md.

The **authoritative source** is Constitution.md in this repository. When this file diverges from Constitution.md, the latter wins. This file exists because:

1. Qwen Code does not always honor @import directives across files; restating the critical rules inline ensures the agent reads them on every session.
2. Cross-agent parity: Claude Code (CLAUDE.md), OpenCode/Cursor (AGENTS.md), Qwen Code (QWEN.md) all start from the same constitutional baseline.

Discover this file via the canonical parent-walk pattern documented in every consuming project's CLAUDE.md / AGENTS.md:

```
find_constitution_root() {  
    local cur="${1:-$PWD}"  
    while [[ "${cur}" != "/" ]]; do  
        if [[ -f "${cur}/constitution/Constitution.md" ]]; then  
            echo "${cur}/constitution"; return 0  
        fi  
        cur="$(dirname "${cur}")"  
    done  
    return 1  
}
```

Identity & posture

When the Qwen Code agent loads this file as part of session bootstrap, it operates under the Helix Constitution with the same identity, anti-bluff posture, and end-user quality covenant as any other CLI agent in the catalogue. There is no “Qwen-lite” interpretation — Qwen Code is held to the full §11.4 covenant.

Top-level invariants for every agent

1. **Read order on a cold start.** `CLAUDE.md` / `AGENTS.md` / `QWEN.md` (this file) for the AI-agent posture; `Constitution.md` for the canonical universal rules; then the consuming project’s `CLAUDE.md` for project-specific extensions.
2. **Multi-mirror push.** Every commit on this submodule **MUST** land on all four canonical mirrors (GitHub + GitLab + GitFlic + GitVerse) per §2.1.
3. **Anti-bluff is non-negotiable.** Passing tests **MUST** imply working features. Bluffing **PASSes** (and **FAILs** that hide root cause) is a release blocker (§11.4 + §11.4.1).
4. **Submodule-catalogue-first.** Before scaffolding anything, survey the `vasic-digital` + `HelixDevelopment` orgs for an existing Submodule (§11.4.74). Reuse → extend → no-match in that order.
5. **Containers-submodule for ALL container workloads.** Use `vasic-digital/containers` (§11.4.76) — never reinvent compose/boot orchestration in-project.

Critical base rules restated (for agents that don’t honour @imports)

§11.4 — Anti-bluff covenant — END-USER QUALITY GUARANTEE

Forensic anchor — verbatim user mandate (2026-04-28, reasserted 2026-05-21, 2026-05-29):

“all existing tests and Challenges do work in anti-bluff manner - they **MUST** confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can’t be used! This **MUST NOT** be the case and execution of tests and Challenges **MUST** guarantee the quality, the completion and full usability by end users of the product!”

Operative rule. Captured tests **MUST** exercise the behavior they claim to verify. **PASS**-without-execution paths, mock-only tests that don’t round-trip, skip-by-default integration tests that mask real failures, metadata-only **PASS**, configuration-only **PASS**, “absence-of-error” **PASS**, and grep-without-runtime **PASS** are all critical defects regardless of how green the summary line looks. Each test failure in CI **MUST** imply a real broken feature; each **PASS** **MUST** imply the feature actually works for the end user. Tests and HelixQA Challenges are bound **EQUALLY**.

The verbatim covenant **MUST** be present in every consumer governance file (project Constitution.md, CLAUDE.md, AGENTS.md, QWEN.md) and in every owned submodule's equivalent. Tools that don't expand @imports still read the literal text.

§1.1 — Mutation-paired gates

Every captured-evidence gate **MUST** ship with a paired mutation test that mutates the gate's anchor + runs the gate + asserts the gate now FAILs. Absence of the paired mutation is itself a bluff.

§11.4.10 — Credentials-handling mandate

Credentials are **NEVER** committed to git, **NEVER** logged, **NEVER** printed to stdout/stderr. .env files are .gitignored; committed siblings are .env.example placeholders only. Pre-store leak audit (§11.4.10.A) scans every PR for credential patterns before merge.

§11.4.17 — Universal-vs-project classification

Every new rule **MUST** declare itself **universal** (applies to every project consuming this Constitution) or **project** (applies only to one specific project). Misclassification (universal rule landing in a project's local CLAUDE.md instead of the constitution submodule) is a release blocker.

§11.4.20 — Subagent-driven-by-default

When a CLI agent (Qwen Code included) has a subagent / Task tool available, the default mode of operation is to dispatch subagents for discrete units of work rather than perform them inline. This protects context, enables parallel work, and matches the §11.4.27 no-fakes-beyond-unit-tests posture (each subagent runs real tools).

§11.4.73 — Main-specification document versioning + revision discipline

Projects with a main specification (docs/specs/.../specification.V<N>.md-shaped) maintain **TWO** version axes:

- **Primary** (V1 / V2 / V3 ...) bumps for major rewrites only.
- **Secondary** (Revision N in the metadata table) bumps for additive changes within a primary version.

Spec edits trigger mandatory comprehensive planning and implementation in the consuming project — they are not isolated doc tweaks (the “spec ripple” rule).

§11.4.74 — Submodule-catalogue-first discovery + extend-don't-reimplement

Direct user mandate (verbatim, 2026-05-20): “We MUST ALWAYS check which already developed features / functionalities do exist as a part of our comprehensive Submodules catalogue located in `vasic-digital` and `HelixDevelopment` organizations on GitHub and GitLab both! Project MUST BE aware of all its existence so we do not implement same things multiple times. For any missing features we MUST IMPLEMENT them properly and extend those Submodules further!”

Before scaffolding any new module / package / helper / utility, the Qwen Code agent MUST: (1) survey `vasic-digital` + `HelixDevelopment` on GitHub + GitLab — the canonical inventory is `submodules-catalogue.md` (142 repos categorised); (2) reuse an existing Submodule when it covers $\geq 80\%$; (3) extend in-place via upstream PR when $80\%+$ matches; (4) document the survey result via `Catalogue-Check: reuse|extend|no-match <org/repo>@<sha>` in the tracker row.

Every Submodule in the catalogue is subject to the same development-cycle rules (§11.4 anti-bluff, §1.1 paired mutations, §11.4.10 credentials, §11.4.61 metadata + ToC, §11.4.65 universal export, §11.4.73 spec versioning, §2.1 multi-mirror push, §3 propagation order).

§11.4.76 — Containers-submodule mandate

Direct user mandate (verbatim, 2026-05-20): “For any work or requirements of running services or codebase inside the Containers (Docker / Podman / Qemu / Emulators, and so on) we MUST USE / INCORPORATE the Containers Submodule properly: `https://github.com/vasic-digital/containers` (`git@github.com:vasic-digital/containers.git`). Containers Submodule contains all means for us to Containerize our code and services! If any feature or Containing System is missing or not supported we MUST EXTEND IT properly like we do all of our projects! No bluff work is allowed of any kind!”

For ANY containerized workload (Docker / Podman / Qemu / Kubernetes / container-backed emulators), the Qwen Code agent MUST: (1) install `vasic-digital/containers` (`digital.vasic.containers`) as a Git submodule when scaffolding the consuming project; (2) consume via `replace` directive during development + pinned commit SHAs in production; (3) boot infra on-demand via the Submodule's `pkg/boot` + `pkg/compose` + `pkg/health` APIs so the operator is never required to start `podman machine` / `docker compose up` manually — the boot is part of the test entry point (**on-demand-infra invariant**); (4) extend the Submodule via upstream PR when a runtime / lifecycle primitive is missing — never reimplement in-project (per §11.4.74); (5) anti-bluff: integration tests claiming to exercise containerized components MUST actually boot them via the Submodule. Short-circuit fakes that bypass boot are a §11.4 violation. A passing test MUST imply the infra was up.

Tracker rows touching containerization MUST record
`Catalogue-Check: extend vasic-digital/containers@<sha>` (or reuse);
`no-match` requires demonstrating the Submodule cannot model the workload.

Planned anti-bluff gate CM-CONTAINERS-USED scans container-touching PRs for `digital.vasic.containers/...` imports. Paired mutation strips the import + asserts FAIL.

Canonical authority: Constitution.md §11.4.76.

Non-compliance: reinventing compose orchestration in-project is a release blocker.

§11.4.77 — Regeneration-mechanism-required mandate

Direct user mandate (verbatim, 2026-05-20): “We must be sure that after excluding anything from Git versioning we still have the mechanism which will out of the box obtain or re-generate missing content! Add this mandatory safety rule / constraint into ours root (constitution Submodule) Constitution.md, CLAUDE.md, AGENTS.md, QWEN.md or any other required document / file from the constitution Submodule!”

Forensic anchor. 2026-05-20T15:00Z: ATMOSphere parent’s audio Tier 1 `commit_all.sh` stalled 4 h on `git add -A` scanning 274 GiB `.git-backup-*` + 159 GiB `RKTools/linux/` + 167 GiB `qa-results/` — all untracked but un-gitignored. Bare `.gitignore` fix would orphan every fresh clone (missing `RKTools`, missing test infra, missing build outputs).

Every `.gitignore` entry the Qwen Code agent considers adding that excludes (a) >~100 MiB OR (b) any artefact essential to building / running / testing the project MUST carry a documented + automated mechanism to either **re-obtain** (vendor tarball, SDK installer, package registry, git submodule, object store) OR **re-generate** (build pipeline, code-gen, asset render, captured-evidence replay, container build).

Required artefacts per qualifying `.gitignore` entry: (1) `.gitignore-meta/<entry-slug>.yaml` declaring pattern + mechanism-type + script-path + expected-disk-usage + vendor-url-or-source + integrity hash + requires-network + requires-credentials; (2) post-clone bootstrap entry (`scripts/setup.sh` or canonical equivalent) running the mechanism non-interactively; (3) pre-build gate verifying regenerated content present OR stamp `.gitignore-meta/.regenerated/<s slug>.ok` recent; (4) README + docs/guides/*.md describing the mechanism + manual fallback + time/disk budget + per-§11.4.10 credentials.

No escape hatch: bare `.gitignore` additions without the mechanism are §11.4 PASS-bluff variants — codebase appears complete but fresh clone cannot build / run. No `--skip-regen-mechanism`, `--gitignore-is-enough`, `--operator-already-has-content` flag.

Planned anti-bluff gate CM-GITIGNORE-REGEN-MECHANISM scans every `.gitignore` addition for matching `.gitignore-meta/` sibling. Paired §1.1 mutation strips a required YAML key → gate FAILs.

Composes with §11.4.6, §11.4.65, §11.4.66, §11.4.71, §11.4.74, §11.4.75, §11.4.76, §9 / §9.2, §3.

Canonical authority: Constitution.md §11.4.77.

Non-compliance is a release blocker regardless of context.

§11.4.78 — CodeGraph code-intelligence mandate

Direct user mandate (verbatim, 2026-05-20): “Make codegraph MANDATORY CHOICE for this purpose for all of our project ... All project which do not have configured and installed codegraph yet MUST DO IT and MUST USE IT!”

Every consuming project worked on by AI coding agents — Qwen Code included — MUST install, initialize, and use **CodeGraph** (<https://github.com/colbymchenry/codegraph>, npm @colbymchenry/codegraph): a local SQLite semantic code-knowledge-graph exposed to agents over MCP, 100% local. Install globally via npm (no sudo). `codegraph init + codegraph index: .codegraph/config.json` is tracked, `.codegraph/codegraph.db` is gitignored with `codegraph index` as its §11.4.77 regeneration mechanism; the `exclude` list MUST exclude other-owned submodules and every §11.4.10 credential/secret path. Wire the `codegraph serve --mcp` server into every CLI agent the developers use — for Qwen Code via `.qwen/settings.json` (`qwen mcp add codegraph codegraph serve --mcp --scope project --transport stdio`). Cover the integration with an anti-bluff verification suite using an unforgeable per-agent challenge (a fact obtainable only by calling a CodeGraph MCP tool); un-runnable agents are documented SKIP gaps, never faked PASSes. Document in `docs/CODEGRAPH.md`. CodeGraph is consumed as the npm package (§11.4.74), not a git submodule.

Composes with §11.4.3, §11.4.10, §11.4.12, §11.4.65, §11.4.30, §11.4.74, §11.4.77, §11.4, §1.1.

Canonical authority: [Constitution.md](#) §11.4.78.

Non-compliance is a process violation; a project worked on by AI agents without CodeGraph installed, wired, and anti-bluff-verified is in breach.

§11.4.79 — Own-org submodules MUST be included in the CodeGraph index

Direct user mandate (verbatim, 2026-05-21): “All Submodules we use in the project and that are part of organizations to which we have the full access via GitHub, GitLab and other CLIs MUST BE included into the codegraph database and initialized / scanned / synced!”

Refines §11.4.78 step 2 with a per-submodule-ownership split. **Own-org submodules** — full write access via the project’s CLIs (canonical orgs: vasic-digital + HelixDevelopment) — MUST be INCLUDED in the index so Qwen Code (and every other AI agent) sees a unified call-graph across the project’s domain + own-org infra code. **Third-party submodules** (the §11.4.74 `no-match → vendor` path) MUST be EXCLUDED. Operational steps: (1) `git submodule update --remote --merge` to pull latest before re-indexing — respect load-bearing third-party pins. (2) Adjust `.codegraph/config.json` `exclude` list. (3) Re-index via `scripts/codegraph_setup.sh`. (4) Validate via `scripts/codegraph_validate.sh` probe resolving a symbol that lives ONLY inside an own-org submodule. (5) Paired §1.1 mutation: temporarily add own-org path to `exclude` → validate FAILs → restore.

Composes with §11.4.74, §11.4.78, §11.4.10, §1.1, §107.

Canonical authority: [Constitution.md](#) §11.4.79.

Non-compliance is a process violation; severe cases (own-org submodules silently excluded without an audit trail in `.codegraph/config.json`) are release blockers.

§11.4.80 — CodeGraph regular-update + sync automation mandate

Direct user mandate (verbatim, 2026-05-21): “We MUST regularly check for the updates and execute codegraph npm updates so the latest version of it is always installed [...] Make sure we have proper full automation bash scripts which will run regularly and that these are part of the constitution Submodule [...] Make sure all updates, sync processes we do and important codegraph related events are all documented under docs/codegraph in Status and Status_Summary documents [...]”

Three deliverables in this constitution submodule: (1) `scripts/codegraph_update.sh` — npm-installs latest; §107 anti-bluff verifies codegraph `--version` reflects the new version. (2) `scripts/codegraph_sync.sh` — runs `status` → `sync` → `status` → project’s validate; appends to both project’s + constitution’s docs/codegraph/Status.md. (3) docs/codegraph/Status.md + Status_Summary.md ledgers, exported per §11.4.65 to `.html` + `.pdf`.

Cadence: weekly floor (§11.4.45). Scripts inherited by reference (§3) — Qwen Code consuming projects invoke them at `${CONST_DIR}/scripts/codegraph_*.sh`, never copy.

Composes with §11.4.78, §11.4.79, §11.4.10, §11.4.45, §11.4.65, §107, §1.1.

Canonical authority: Constitution.md §11.4.80.

Non-compliance is a process violation; severe cases (>2 weeks since last update + open AI-agent work) are release blockers.

§11.4.81 — Cross-platform-parity mandate (User mandate, 2026-05-21)

Direct user mandate (verbatim, 2026-05-21): “Any Linux-only blocker / issue we have MUST BE created macOS and other supported platforms equivalent! So, depending on platform proper implementation will be used for particular OS! EVERYTHING MUST BE PROPERLY EXTENDED AND UPDATED!”

Every consuming project whose supported-platforms manifest lists more than one OS MUST ship per-OS-equivalent implementations + tests for every feature/gate/challenge/mutation that depends on platform-specific primitives. Runtime dispatch via `uname -s` (or equivalent platform detection). Three sub-mandates: **(A)** Per-OS implementation REQUIRED — `cgroup/systemd/proc` primitives have documented equivalents (POSIX `setrlimit/ulimit`, `launchd`, `rctl`, Job Object). **(B)** Per-OS tests REQUIRED — every gate test has case `"$(uname -s)"` in branches with positive captured evidence per §11.4.2 + §11.4.5 in each branch; SKIP-with-reason only when the platform genuinely cannot enforce. **(C)** Honest kernel-gap citation + adjacent equivalent test REQUIRED where no equivalent exists (canonical: XNU does NOT enforce `RLIMIT_AS` for unprivileged processes → SKIP with exact reproducer + adjacent test of what IS enforced, e.g. `RLIMIT_CPU+SIGXCPU`). The adjacent test is itself anti-bluff per §11.4 with a paired §1.1 mutation.

Per-OS equivalence catalogue (canonical): systemd-run --user --scope [?] POSIX ulimit -t -u / launchd; cgroup MemoryMax [?] XNU gap (use RLIMIT_CPU adjacent) / rctl / Job Object; cgroup TasksMax [?] RLIMIT_NPROC; / proc/<pid>/oom_score_adj [?] no Darwin/BSD equivalent.

Composes with §11.4.1 / §11.4.2 / §11.4.3 (strictened — SKIP only when kernel cannot) / §11.4.4 / §11.4.5 / §11.4.6 / §11.4.20 / §11.4.27 / §11.4.69 / §11.4.70 / §107. Pre-build gate CM-CROSS-PLATFORM-PARITY + paired §1.1 mutation. No escape hatch.

Canonical authority: Constitution.md §11.4.81.

Non-compliance is a release blocker on multi-platform projects.

§11.4.82 — Iteration-speedup discipline mandate (User mandate, 2026-05-22)

Direct user mandate (verbatim, 2026-05-22): “How can we speed-up this whole development and fixing process? ... all speed optimizations critical rules and mandatory constraints MUST BE all added into our root (constitution Submodule) Constitution.md, CLAUDE.md, AGENTS.md and QWEN.md!”

Iteration cycle time bounds the project’s defect-discovery rate. Slow cycles ship fewer validated features per unit of operator time. The 2026-05-22 forensic session witnessed ~3 hours of operator wait on a job whose intrinsic compute was ~90 min — every wasted minute came from a missing speedup discipline.

Every consuming project’s build / test / commit / debug pipeline MUST adopt the following 9 speedup disciplines AS MANDATORY:

- **(A) Phase 1 forensic before any speculative source patch.** Speculative patches without root-cause evidence are §11.4.6 + §11.4.82 violations.
- **(B) Live-ADB-First (or live-equivalent) before any rebuild.** §11.4.51 strengthened to release-blocker. Skipping live-probe for LIVE_ADB_TESTABLE changes = §11.4.82 violation.
- **(C) Pre-flight before rebuild orchestrators.** 30-sec readiness check verifies device + sink reachability, memory budget, lock files, orphan processes.
- **(D) Persistent build caches outside containers.** ccache + Soong / sccache / Gradle daemon state bind-mounted to host. Subsequent rebuilds drop from ~40 min to 5-15 min.
- **(E) Module-only rebuild for CONFIG_*=m driver patches.** make modules on single driver dir saves 15-20 min/cycle when applicable.
- **(F) Parallel multi-device testing.** Every owned validation device runs the autonomous cycle concurrently with separate output dirs. Catches per-device defects one cycle earlier.
- **(G) Subagent scope ≤30 min + worktree isolation + single-responsibility.** Empirical pattern: 8-task subagents stall, 2-3-task subagents complete.
- **(H) Lock-file + stale-process hygiene.** Clean .git/index.lock + orphan processes on session start. Disable auto git-gc in concurrent multi-agent repos.
- **(I) Cycle telemetry per §11.4.24.** Every cycle logs commit, per-phase wall-clock, speedup flag set, outcome. Weekly aggregation surfaces empirical ROI per discipline.

Composes with §11.4.4 / §11.4.6 / §11.4.9 / §11.4.20 / §11.4.24 / §11.4.42 / §11.4.43 / §11.4.50 / §11.4.51 / §11.4.52 / §11.4.58 / §11.4.70 / §12.7 / §107. Pre-build gate CM-ITERATION-SPEEDUP-DISCIPLINE (when implemented) + paired §1.1 mutation. No escape hatch — no `--skip-phase1`, `--no-pre-flight`, `--rebuild-everything`, `--unlimited-subagent-scope`, `--ignore-locks`, `--no-telemetry` flag exists.

Canonical authority: Constitution.md §11.4.82.

Non-compliance is a release blocker.

§11.4.83 — docs/qa/ end-user evidence mandate (User mandate, 2026-05-22)

Direct user mandate (verbatim, 2026-05-22): “every feature that ships MUST carry a recorded e2e communication transcript + any attached materials under docs/qa/<run-id>/ (per-feature subdirectories). A feature with no QA transcript is itself a §107 PASS-bluff — it claims to work but has no auditable runtime evidence. Bot-driven automation MUST preserve full bidirectional communication threads as proof.”

Every feature that ships MUST carry a recorded end-to-end communication transcript plus attached materials committed under docs/qa/<run-id>/. Transcripts MUST be full bidirectional. Attached materials live in-repo (no external-only links — §11.4.13 sink-side violation). Bot-driven / agent-driven QA MUST preserve the full conversation thread as the proof artefact. CI release gates MUST refuse to tag a version whose feature-shipping commit lacks its docs/qa/<run-id>/.

Composes with §11.4.2, §11.4.5, §11.4.13, §11.4.65, §11.4.69, §107, §1.1.

Canonical authority: Constitution.md §11.4.83.

Non-compliance is a release blocker.

§11.4.84 — Working-tree quiescence rule for subagent commits (User mandate, 2026-05-22)

Short tag: working-tree quiescence.

Direct user mandate (verbatim, 2026-05-22): “no subagent commit may proceed while any concurrent mutation gate is in flight in the same checkout. Before `git add`, the committing agent MUST `grep` its own working tree for mutation markers (MUTATED for paired, // always pass, return `json.Marshal` shortcut paths, etc.). Any unexplained file in the staging area triggers ABORT.”

No subagent commit may proceed while any concurrent mutation gate is in flight in the same checkout. Pre-`git add`: `grep` for mutation markers (MUTATED for paired, // always pass, return `json.Marshal` shortcuts, // MUTATION annotations, `_mutated_*` filenames). Cross-check `git status --porcelain` against declared scope → unaccounted entries ABORT.

Lesson (forensic). A consuming project's logo-fix subagent (Herald commit 72e81ab, 2026-05-21) swept a // always pass JWT-bypass mutation residue into an unrelated commit, pushed to all four mirrors before being caught. Fix d5bd360 landed within the hour, but the security-defect window is real and the lesson is constitutional.

Operative rule. (1) Pre-git add grep + scope-match → unaccounted ABORT. (2) Active mutation gates MUST be serialised before unrelated commits. (3) Concurrent subagents in same checkout MUST use lockfile (.git/MUTATION_IN_PROGRESS) or git worktree add. (4) Pre-push mutation-residue-scanner BLOCKS pushes containing mutation markers.

Composes with §1.1, §11.4.20, §11.4.70, §11.4.27, §11.4.10, §11.4.71, §107.

Canonical authority: Constitution.md §11.4.84.

Non-compliance is a release blocker.

§11.4.85 — Stress + Chaos Test Mandate (User mandate, 2026-05-24)

Short tag: stress-chaos-mandate.

Direct user mandate (verbatim, 2026-05-24): “Every fix or improvement you do MUST BE covered with full automation stress and chaos tests so we are sure nothing can break the functionality and all edge cases are monitored and polished and additionally fixed if that is needed! Everything must produce rock solid proofs and follow fully no-bluff policy!”

Every fix MUST ship with full-automation stress + chaos test suites. Happy-path-only coverage is a §11.4 / §107 PASS-bluff at the resilience layer.

Stress = sustained load ($N \geq 100$ iters OR ≥ 30 s) + concurrent contention ($N \geq 10$ parallel) + boundary conditions (empty/max/off-by-one). **Chaos** = failure-injection per §11.4.69 closed-set (process-kill, network-drop, input-corruption, OOM, disk-full, state-corruption) + verifiable recovery.

Anti-bluff: every stress + chaos PASS cites a captured-evidence artefact (latency.json / categorised_errors.txt / state_delta_snapshot.json) per §11.4.5 + §11.4.69. Helper library stress_chaos.sh provides ab_stress_run / ab_stress_concurrent / ab_chaos_* primitives. Chaos cleanup non-negotiable — corrupt-restore / disk-fill-cleanup / process-restart MUST run in trap EXIT.

4-layer per §11.4.4(b): pre-build gate (test files exist + parse) + paired meta-test mutation + on-device test (if LIVE_ADB_TESTABLE) + HelixQA Challenge entry. Composes with §11.4 / §11.4.1 / §11.4.5 / §11.4.6 / §11.4.43 / §11.4.50 / §11.4.52 / §11.4.69 / §11.4.83.

Canonical authority: Constitution.md §11.4.85.

Non-compliance is a release blocker. No --skip-stress, --no-chaos, --happy-path-suffices flag exists.

§11.4.86 — Roster/corpus-backed Status-doc auto-sync mandate (User mandate, 2026-05-25)

Forensic anchor (verbatim): “Make sure that assets and players Status docs are ALWAYS regularly updated and in sync like all others Status docs — any time we add or modify the assets content(s) or we change or add new / remove existing pre-installed video and audio player apps! This MUST WORK OUT OF THE BOX!”

Status docs (§11.4.45) backed by a **tracked roster** (installed apps/components) or **asset corpus** (test/media directory) MUST resync out of the box the moment a member changes — Status doc + Status_Summary + HTML + PDF, mechanically. Mechanism (all must hold): (1) drift-proof **fingerprint** = sha256 of sorted member list (NOT mtime), persisted in a sidecar; (2) **sync helper** regenerates fingerprint + re-exports (via §11.4.65); (3) **pre-build gate** FAILs on fingerprint drift (mirrors §11.4.12 + §11.4.45); (4) paired §1.1 mutation. Composes with §11.4.12 / §11.4.45 / §11.4.53 / §11.4.56 / §11.4.57 / §11.4.59 / §11.4.60 / §11.4.65 / §11.4.6. Universal (§11.4.17) — consuming project supplies the specific docs/roster/helper/gate per §11.4.35.

Canonical authority: Constitution.md §11.4.86.

Non-compliance is a release blocker. No --skip-roster-sync, --allow-status-drift flag exists.

§11.4.87 — Endless-loop autonomous work + zero-idle agent dispatch + anti-bluff testing mandate (User mandate, 2026-05-26)

When operator instructs an AI agent to “continue in endless loop fully autonomously” (or semantically-equivalent), the agent MUST treat as HARD-CONTRACT covenant: (A) continue until docs/Issues.md non-terminal Status entries = 0 AND docs/CONTINUATION.md §3 Active work empty AND no subagent in-flight AND no external dep in-flight; (B) dispatch background subagents for parallelisable work — main + subagents concurrent, “waiting for results” is the ONLY idle reason; (C) every closure lands four-layer test coverage per §11.4.4(b) with captured-evidence “physical proofs” (audio/video/network/UI/sysfs) — metadata-only / config-only / absence-of-error / grep-without-runtime PASS are critical defects; (D) §11.4 anti-bluff covenant family operative end-to-end (tests AND HelixQA Challenges bound equally per forensic anchor “tests pass but features don’t work”); (E) loop terminates ONLY on all-conditions-met, explicit operator STOP, host-safety demand, or scheduled wake on known-future-actionable signal.

Composes with §11.4 / §11.4.1 / §11.4.2 / §11.4.4 / §11.4.5 / §11.4.6 / §11.4.7 / §11.4.20 / §11.4.27 / §11.4.42 / §11.4.43 / §11.4.50 / §11.4.52 / §11.4.58 / §11.4.68 / §11.4.69 / §11.4.70 / §11.4.83 / §11.4.85 / §11.4.86 / §12.10. Pre-build gate CM-COVENANT-114-87-PROPAGATION + paired §1.1 mutation.

Canonical authority: Constitution.md §11.4.87.

Non-compliance is a release blocker. No --idle-OK, --skip-endless-loop, --bluff-permitted-for-this-task, --metadata-only-test-suffices, --no-physical-proof-required flag exists.

Companion documents

- Constitution.md — authoritative universal Constitution. ALWAYS the tie-breaker.
- CLAUDE.md — Claude Code agent's view of the universal constitution.
- AGENTS.md — OpenCode / Cursor / generic-tooling view.
- README.md — high-level overview + multi-mirror contract.

When operating in a consuming project, ALSO read the project's local QWEN.md / CLAUDE.md / AGENTS.md for project-specific extensions; these MUST NOT weaken any universal rule but MAY add stricter project-specific constraints.

§11.4.88 — Background-push mandate: commit-lock release immediately after commit, push runs detached (User mandate, 2026-05-26)

Forensic anchor: 2026-05-26 commit_all.sh held flock ~5h on synchronous post-commit push. Mandate: (A) flock release IMMEDIATELY after commit; (B) push detached via nohup + disown; (C) push_all.sh per-remote flock — same-remote serializes, different-remote parallel; (D) failures → qa-results/push_failures/, autonomous loop checks per §11.4.87(A); (E) --sync-push escape for §11.4.41 force-push only. Composes §2.1 / §9.2 / §11.4.41 / §11.4.42 / §11.4.71 / §11.4.87. Gates CM-COVENANT-114-88-PROPAGATION + CM-BACKGROUND-PUSH-WIRED + paired §1.1 mutations.

Canonical authority: Constitution.md §11.4.88. Non-compliance is a release blocker.

§11.4.89 — Background test execution mandate (User mandate, 2026-05-27)

Forensic anchor 2026-05-27: conductor invoked long pre_build synchronously, blocking main stream 6-7 min on §JV/§JW/§JX/§JY scaffolding. Symmetric to §11.4.88 at test-execution layer. Mandate: (A) long tests (>30s) run via nohup ... > <log> 2>&1 & + disown; (B) main stream proceeds to §11.4.42 priority queue immediately; (C) hard-dependency gating via poll/pgrep — surface §11.4.66 options if still running; (D) failures land in log files; (E) foreground only <30s OR operator authorisation; (F) per-script flock. Composes §11.4.42 / §11.4.66 / §11.4.82 / §11.4.84 / §11.4.85 / §11.4.87 / §11.4.88. Gates CM-COVENANT-114-89-PROPAGATION + CM-BACKGROUND-TEST-EXECUTION-WIRED + paired §1.1 mutations.

Canonical authority: Constitution.md §11.4.89. Non-compliance is a release blocker.

§11.4.90 — Obsolete status + obsolescence audit (User mandate, 2026-05-27)

§11.4.15 Status closed-set + terminal Obsolete (→ Fixed.md). Reasons: superseded-by-design-change | superseded-by-later-mandate | feature-removed | duplicate-of | unsupported-topology.*Obsolete-

Details:** line mandatory (Since + Reason + Superseding-item + Triple-check evidence). Colorizer cell-status-obsolete = light-gray + strikethrough. Audit cadence per §11.4.40 + §11.4.42. Triple-check non-negotiable.

Canonical authority: Constitution.md §11.4.90. Release blocker.

§11.4.91 — Summary-doc clarity (User mandate, 2026-05-27)

Every summary entry's description: self-contained, meaningful, ≥ 6 words / ≥ 40 chars, SUBJECT + PROBLEM/GOAL. Anti-patterns forbidden: section labels, bare metadata, §-letter alone. Generators extract from H1/H2 heading. Pre-build gate CM-SUMMARY-CLARITY-DESCRIPTIONS.

Canonical authority: Constitution.md §11.4.91. Release blocker.

§11.4.92 — Multi-pass change-evaluation (User mandate, 2026-05-27)

5-pass evaluation BEFORE commit-ready: (1) Main-task verification; (2) Regression-blast-radius; (3) Cross-feature interaction; (4) Deep-research per §11.4.8; (5) Anti-bluff confirmation. Doc per pass. Trivial exemption: typo/revision/MD-export ONLY if zero source touched.

Canonical authority: Constitution.md §11.4.92. Release blocker.

§11.4.93 — SQLite-SSoT for workable items (User mandate, 2026-05-27)

docs/.workable_items.db SQLite — DB authoritative, MD derived. Schema: items + item_history + obsolete_details + operator_block_details + firebase_metadata + meta. Go binary cmd/workable-items/ sync md db + diff + validate + add + close. Bidirectional regen byte-identical. Anti-bluff: unit+integration+stress+chaos per §11.4.85. Go binary in constitution submodule per §11.4.74. 6-phase migration §LA.

Canonical authority: Constitution.md §11.4.93. Release blocker.

§11.4.94 — Zero-idle priority-first parallel-by-default (User mandate, 2026-05-27)

Binds §11.4.20/42/58/70/72/82/87/88/89 — always-on. Idle ONLY: externally blocked, operator STOP, §12 host-safety. Before any wake/sleep survey parallel-work queue. Priority MANDATORY (§11.4.72 audio first). Subagent-driven default non-trivial. Background default >30s. Stability per §11.4.92 + §11.4.84 + §12.6-9. Updates at milestones.

Canonical authority: Constitution.md §11.4.94. Release blocker.

§11.4.95 — Workable-items DB TRACKED in git (User mandate, 2026-05-27)

§11.4.93 amendment: docs/workable_items.db TRACKED, NEVER gitignored. Authoritative source. Every mutation → stage+commit+push DB + MD. WAL-checkpoint before commit. §11.4.77 does NOT apply. Pre-build gates CM-COVENANT-114-95-PROPAGATION + CM-WORKABLE-ITEMS-DB-TRACKED.

Canonical authority: Constitution.md §11.4.95. Release blocker.

§11.4.96 — Safe-parallel-work-with-long-build (User mandate, 2026-05-27)

SAFE during AOSP build: (A) docs/MD; (B) scripts/; (C) pre-build gates; (D) on-device tests; (E) constitution edits+push; (F) submodule push per §11.4.88; (G) read-only ADB probes; (H) subagent dispatch; (I) web research; (J) workable-items DB ops; (K) pre-build + meta-test execution backgrounded.

UNSAFE: (α) git checkout/reset on source; (β) mass deletes/renames under built source; (γ) APK-built submodule pointer; (δ) out/; (ε) make clean; (ζ) container destruction; (η) disk-fill; (θ) §12 host-safety.

Conductor dispatches ≥ 1 (A)-(K) per pause. “Build running, nothing else to do” NEVER true.

Canonical authority: Constitution.md §11.4.96. Release blocker.

§11.4.97 — Max-use-of-idle + progress-update cadence (User mandate, 2026-05-27)

1. Every minute progressable idle = §11.4.97 violation, dispatch CONTINUOUSLY; (B) 1-line operator updates at commit/subagent/anchor/evidence/migration — no prompt; (C) continuous physical-proof gathering per §11.4.5/6/69; (D) composes with operating-mode anchor family; (E) idle-only-when-blocked closed-set unchanged from §11.4.94(A). Pre-build gates CM-COVENANT-114-97-PROPAGATION + CM-IDLE-TIME-AUDIT.

Canonical authority: Constitution.md §11.4.97. Release blocker.

§11.4.98 — Full-Automation Anti-Bluff (User mandate, 2026-05-28)

Forensic anchor: “Make sure we have full automation testing of all scenarios with real bot, main group and users without any manual intervention or contribution of real user! ... No bluff is allowed!”

Closes the manual-intervention gap §11.4+85+87+89+94 didn’t forbid. (A) every test MUST self-drive end-to-end — PASS/FAIL/SKIP-with-reason without further human action; (B) single exception = one-time credential bootstrap OUTSIDE test execution (.env, ~/.bashrc, OAuth, MTPProto-session-activation); (C) concrete live requirements: (1) no “operator MUST type” prompts — drive via MTPProto/real-user-API/webhook-fixture/in-process loopback; (2) no UUIDs colliding with active dev session (Herald 2026-05-28: silent exit -1 lesson); (3) no 60s human-response

windows (§11.4.50 violation); (4) re-runnability -count=3 with self-cleaning state; (5) audit COMPLIANT vs NON-COMPLIANT; (6) no false-positive PASS — silent-skip-as-PASS + stale-evidence forbidden, §11.4.3 SKIP-with-reason correct; (D) composes §11.4.85+89+87+94 = continuously-validated fully-automated non-flake anti-bluff regime; (E) inheritance per §11.4.35 — restate literal 11.4.98 in every consumer; pre-build gate CM-COVENANT-114-98-PROPAGATION; paired §1.1 mutations strip → FAIL; (F) enforcement: manual-action commit BLOCKED; NON-COMPLIANT after 30 days → §11.4.90 Obsolete citing §11.4.98.

Canonical authority: Constitution.md §11.4.98. Release blocker.

§11.4.99 — Latest-Source Documentation Cross-Reference (User mandate, 2026-05-28)

Forensic: “ALWAYS check against latest versions of services we use web/online docs before creating instructions! ... mandatory rules / constraints, result is consistency and safety of created instructions, guides and manuals!”

Case study (Herald 2026-05-28): MTProto guide recommended VoIP + omitted recover@telegram.org pre-login email; contradicted official Telegram + gotd/td docs; could have caused permanent ban. Misguidance-by-stale-docs = §11.4 PASS-bluff at documentation layer.

1. Pre-commit cross-reference: fetch LATEST official online docs (WebFetch/MCP/direct browsing — NEVER training data) for every operator-facing instruction doc; cite source URL+date in “## Sources verified” footer AND commit-message footer; seek secondary authoritative sources for sparse official docs. (B) Negative findings documented explicitly. (C) STALE after 6 months (90 days for risk-classified services); re-verify at vN.0.0, on breaking-change, on operator-error. (D) Risk-classified: messengers/cloud/payment/AI/code-hosting/package-managers. (E) Composes with §11.4.92 Pass 4 INDEPENDENT — cannot substitute. (F) Inheritance per §11.4.35 — restate literal 11.4.99 in every consumer’s CLAUDE/AGENTS/QWEN. (G) Enforcement: missing-footer BLOCKED; stale-beyond-grace → §11.4.90 Obsolete.

Canonical authority: Constitution.md §11.4.99. Release blocker.

§11.4.100 — RETIRED. Demoted to consumer project (ATMOSphere video-color/visual-quality fidelity) per §11.4.17/§11.4.35 — project-specific (RK3588/MPV/Arvus), not universal. See the consuming project’s Constitution/CLAUDE/AGENTS/QWEN.

§11.4.101 — Autonomous-decision-over-blocking (User mandate, 2026-05-28)

Forensic: “when working in endless working loop fully autonomously try to decide most properly about points which would block execution and wait for us. If we haven’t answered now work would be blocked whole night! If possible and if that will not cause any issues make proper and most reliable and safe decision so we achieve maximal efficiency and work gets fully done!”

In autonomous / endless-loop mode (per §11.4.87) the agent **MUST** minimize operator-blocking and make the safe, reliable, reversible decision itself — work NOT stalled overnight waiting for input. §11.4.87 = keep working; §11.4.101 = HOW to clear the decision points.

Decision rule (proceed autonomously when ALL): (a) reversible OR pre-op backup per §9.2; (b) safe choice determinable from captured evidence per §11.4.6 (LIKELY ≠ determination); (c) wrong-choice blast radius bounded AND recoverable; (d) composes with §11.4 + §12 + §9. Block-only-when (BLOCK via §11.4.66 ONLY when ALL): irreversible AND high-blast-radius AND choice undeterminable from evidence — external-account state, inaccessible hardware, destructive-op-without-backup, force-push (§9.2 + §11.4.41), spending / third-party send. Operator-blocked per §11.4.21 only after this fires + self-resolution-exhaustion audit. Maximize-progress-while-blocked: a block parks one work unit, not the loop — keep progressing NON-blocked items per §11.4.87 + §11.4.94; pose-and-idle = §11.4.94 + §11.4.97 violation. Composes §11.4.6/21/40/41/66/87/94/§9.2/§12. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-101-PROPAGATION (literal 11.4.101) + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: Constitution.md §11.4.101. Release blocker. No --always-block-on-decision / --never-decide-autonomously / --skip-decision-rule / --block-without-self-resolution escape.

§11.4.102 — Systematic-debugging always-on + always-loaded skill-discovery + plugin availability (User mandate, 2026-05-29)

Forensic: “ALWAYS trigger / start the”/superpowers:systematic-debugging” skills when any issues happen! ... we **MUST** activate the skill(s) and make strongest efforts in full in depth analysis / debugging and determine root causes ... “/using-superpowers” skill is ALWAYS loaded, applied and used! All dependencies (plugins) that Claude Code or other market places are offering **MUST BE** installed if these are not already available for loading and use!”

1. On ANY spotted issue/bug/test-failure/gate-failure/regression/misalignment/inconsistency/unexpected-behaviour the agent **MUST** activate superpowers:systematic-debugging (or platform-equivalent) BEFORE proposing/writing/applying any fix — Iron Law: NO FIXES WITHOUT ROOT CAUSE INVESTIGATION FIRST. Four-phase arc: root-cause → pattern → hypothesis (prove/disprove with captured evidence) → implementation (against proven cause only). Guess-and-retry / symptom-patch / re-run-hoping-it-passes (“probably transient/flaky”) without completed investigation = §11.4.102 violation; calling a failure transient/flaky/intermittent without captured forensic evidence = simultaneous §11.4.6 + §11.4.7 violation. (B) superpowers:using-superpowers (or platform-equivalent skill-discovery) ALWAYS loaded + applied at session start, consulted before any task — if ANY skill could apply (even 1%) it **MUST** be invoked, never improvised. (C) Every mandated skill plugin/dependency (Claude Code marketplaces or other runtime ecosystems) **MUST** be installed + loadable BEFORE dependent work; missing plugin blocking a mandated skill = release-blocker. Install: runtime’s own path — Claude Code /plugin flow (add marketplace → install → confirm skill in available-skills list); other runtimes’ documented installer. Anti-bluff: confirm via live capability list (install exit 0 ≠ skill loadable, §11.4.80 lesson). Composes §11.4.4/6/7/8/43/70/82(A)/92. Classification: universal (§11.4.17).

Propagation gate CM-COVENANT-114-102-PROPAGATION (literal 11.4.102) + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: Constitution.md §11.4.102. Release blocker. No --skip-systematic-debugging / --guess-and-retry-OK / --symptom-patch-permitted / --skip-skill-discovery / --plugin-optional / --missing-plugin-is-warning escape.

§11.4.103 — Continuous parallel-stream working routine (User mandate, 2026-05-29)

Forensic: “Do this working approach continuously and make it part of regular working routine and add it to the Constitution Submodule documented fully.”

Promotes the proven multi-stream pattern into the project’s standing default working routine (not a per-request opt-in). Load-bearing invariant: main work stream MUST always stay FREE. (A) Main stream FREE — ALL commit AND push run detached (nohup ... & + disown per §11.4.88), never blocking on push or slow mirror. (B) ≥ 3 parallel background streams at all times + auto-backfill (User mandate 2026-05-29; raised from ≥ 2) — at least three subagent-driven streams (§11.4.70/§11.4.20, isolated §11.4.58 + §11.4.84) alongside main whenever three-plus non-contending actionable items exist; the moment any one stream is FULLY done a new stream MUST immediately start + take its place (next-highest-priority non-contending item), count NEVER drops below 3 while actionable items remain. Idle below 3 only when no remaining items OR all externally blocked (§11.4.94/97/101). Band 3–6, bounded §12.6 60% mem + §11.4.58 6-agent cap. (C) Most-critical + most-visible first; audio always top §11.4.72 + §11.4.42. (D) Safe-during-build scope only — while heavy gradle/m -j/42 GB containerised AOSP build (§12.9) runs, streams restrict to §11.4.96 SAFE catalogue; NEVER a second concurrent heavy build (§12.8). (E) Heavy anti-bluff every closure (§11.4.5/6/50/69/102; root cause proven or UNCONFIRMED:/UNKNOWN:/PENDING_FORENSICS:). (F) Idle ONLY when genuinely externally blocked (hardware/network/in-flight build-test-push) OR operator STOP OR §12 host-safety per §11.4.94(A)+§11.4.97; block parks one work unit not the loop (§11.4.101). Composes §11.4.58/70/72/87/88/89/94/96/97/101/102/42/84/§12.6-§12.9/§9.2. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-103-PROPAGATION (literal 11.4.103) + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: Constitution.md §11.4.103. Release blocker. No --block-main-stream / --synchronous-commit / --synchronous-push / --single-stream-only / --skip-parallel-streams / --serialise-actionable-work / --idle-without-queue-survey escape.

§11.4.104 — Participant identity, attribution & notification-tagging (User mandate, 2026-05-31)

Forensic: “Every supported messenger must relate messages to participants (Subscribers/Users); the same logical person may have a different username on every messenger. Workable items must carry who created them and who they are assigned to. Notifications must @-tag the right participant — but never the operator (who drives the system) and never the system agent.”

MANDATORY for every consumer shipping a messenger/notification surface (Herald + flavor binaries = reference impl; others inherit §11.4.35). Detailed spec: Herald docs/design/PARTICIPANT_ATTRIBUTION.md — restated, not redefined. (A) Every messenger MUST relate messages to a **Participant** (logical Subscriber/User); SAME person MAY have a DIFFERENT username per messenger (logical subscriber: canonical messenger-neutral handle, kind ∈ {human, agent, service}; + per-channel aliases channel/channel_user_id/@username for tagging). Canonical handle closed set: Claude (reserved system-agent sentinel; never tagged) OR a subscriber @username. (B) Operator = env var HERALD_<CHANNEL>_OPERATOR_USERNAME, not a DB flag — the one human who drives via the agent CLI; a normal Participant whose handle equals that value. (C) Workable items MUST carry created_by + assigned_to (canonical handles): CLI-prompt→Operator; system/agent-detected→Claude; received-message→sender's resolved @username; assigned_to defaults to Operator, overridable; legacy items carry "" and MUST still parse + validate. (D) Tagging matrix: tag assigned_to if human ≠ Operator; tag created_by if human ≠ Operator ≠ Claude; NEVER tag Claude or the Operator; de-dup; resolve to channel @username, skip if absent there. (E) Anti-bluff (composes §11.4): real SQLite round-trip with the new columns byte-identical (incl. legacy fixtures WITHOUT the fields); tagging matrix proven by a truth-table + cell-flip mutation forcing FAIL; E2E real-event → real-message asserting the exact @usernames + a NEGATIVE case proving the Operator is NOT tagged; evidence under docs/qa/<run-id>/. Composes §11.4 + §11.4.1..16 / §11.4.5 / §11.4.69 / §11.4.50 / §11.4.91 / §11.4.93 / §11.4.95 / §1.1. Classification: universal (§11.4.17); no-messenger projects inherit latently (§11.4.96 pattern). Propagation gate CM-COVENANT-114-104-PROPAGATION (literal 11.4.104) + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: Constitution.md §11.4.104; detailed spec Herald docs/design/PARTICIPANT_ATTRIBUTION.md. Release blocker. No --skip-attribution / --no-participant-tagging / --tag-operator-anyway / --attribution-later escape.

§11.4.105 — Natural-language intent recognition & clarification (User mandate, 2026-05-31)

Forensic: “Users must NOT need to know command syntax (no **COMMAND:** ... prefix). They send a clear natural-language message; the System determines the intent. The System recognizes the commands it has; if none matches it infers the exact intent; if it is totally unable it replies, tags the user (@user ...), and asks to clarify precisely. We MUST always do our best to determine exact intent so we never annoy end users. This is a CORE part of the System.”

MANDATORY for every consumer shipping a messenger/command surface (Herald + flavor binaries = reference impl; others inherit §11.4.35). Detailed spec: Herald docs/design/INTENT_RECOGNITION.md — restated, not redefined. (A) No required command syntax — users MUST NOT need to know any syntax (no **COMMAND:** prefix); they send plain natural language, the System determines the intent. (B) Three-tier resolution, first that succeeds wins: TIER 1 recognize the System's existing command set from natural language → action (confident deterministic match); TIER 2 when no command matches, infer the exact intent (LLM maps language → action), NEVER guessing; TIER 3 when neither a command nor a confident intent is determinable, REPLY to the message, TAG the sender (@username, via the §11.4.104 IdentityResolver) and ask a PRECISE clarifying question NAMING the candidate intents — no guessing, no silent drop.

(C) Never guess, never drop — a wrong action is worse than a clarifying question (composes §11.4.6); a message is NEVER silently dropped; only genuine ambiguity reaches Tier 3, which always replies-tags-and-asks. (D) Anti-bluff (composes §11.4): every tier ships unit + integration + E2E + full-automation tests with real captured evidence — Tier 1 truth-table (natural-language → action+fields + conservative negatives that MUST fall through to “no match”); Tier 3 E2E whose recorded reply body is EXACTLY @<sender> <specific question> + a NEGATIVE proving a clear command does NOT trigger clarify; a paired §1.1 mutation breaking the confidence guard (false-match) OR dropping the clarify tag MUST FAIL a test; evidence under docs/qa/<run-id>/. Composes §11.4 + §11.4.1..16 / §11.4.6 / §11.4.104 / §11.4.5 / §11.4.69 / §11.4.98 / §1.1. Classification: universal (§11.4.17); no-messenger projects inherit latently (§11.4.96 pattern). Propagation gate CM-COVENANT-114-105-PROPAGATION (literal 11.4.105) + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: Constitution.md §11.4.105; detailed spec Herald docs/design/INTENT_RECOGNITION.md. Release blocker. No --require-command-syntax / --guess-intent-ok / --skip-clarify / --drop-on-ambiguous escape.

§11.4.106 — Docs Chain — mechanical documentation/DB sync engine (Operator mandate, 2026-05-31)

Forensic: Docs Chain is the canonical mechanical enforcer of the documentation-sync mandates; consumers MUST use the engine instead of ad-hoc per-project doc-sync scripts, register chains via per-context YAML, and never accept a faked transform.

MANDATORY for every consumer. Docs Chain (the vasic-digital/docs_chain engine — a universal Go bidirectional document-and-database dependency-propagation engine) is the canonical mechanical enforcer of the documentation-sync mandates. Detailed spec: engine docs ~/Projects/docs_chain → docs/CONSTITUTION_INTEGRATION.md (distribution + inheritance + anchor mapping table) + docs/USE_CASE_CATALOGUE.md (chain recipes) — restated, not redefined. (A) Use the engine, never ad-hoc scripts — consumed by reference (the flat-layout sibling ~/Projects/docs_chain / the constitution-exposed path), inherited like §11.4.80’s codegraph_* scripts: referenced, NEVER copied; ad-hoc sync_*/generate_*/summary/update_readme_doc_links scripts superseded + retired per registered context. (B) Consumer-owned contexts — the engine is project-agnostic; the consumer registers chains as data via .docs_chain/contexts/*.yaml (§11.4.28 decoupling); state.json + *.docs_chain.tmp gitignored. (C) Anchors mechanized (per CONSTITUTION_INTEGRATION mapping table): §11.4.12/.53/.45/.56/.57/.59/.60/.65/.86/.93/.95/.44 + §12.10 — content-hash change detection (NOT mtime, §11.4.86), atomic-rename + SQLite-txn commit + rollback (§9.2), both-dirty sync → conflict-not-silent-merge (§11.4.6), verify as the deterministic CI/pre-build gate (§11.4.50), per-run captured evidence to qa-results/docs_chain/<run-id>/ (§11.4.69). (D) NOT a replacement for authoring discipline — the source author still writes the §11.4.44 revision header; the engine only keeps exports in sync. (E) Anti-bluff (composes §11.4): NEVER fakes a transform — a missing pandoc/weasyprint surfaces a typed ToolAbsentError + honest §11.4.3 SKIP-with-reason, never a fake PASS / partial write; every sync/verify carries real captured evidence. Composes §11.4 + §11.4.1..16 / §11.4.6 / §11.4.28 / §11.4.80 / §9.2 / §11.4.50 / §11.4.69 / §11.4.5 / §1.1 + the mechanized sync anchors. Classification: universal (§11.4.17); no derived-

export/DB-sync projects inherit latently (§11.4.96 pattern). Status (§11.4.6): engine Phases 1–3 IMPLEMENTED+tested; CLI/YAML loader (Phase 4) + submodule distribution (Phase 6) PLANNED + OPERATOR-GATED. Propagation gate CM-COVENANT-114-106-PROPAGATION (literal 11.4.106) + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: Constitution.md §11.4.106; detailed spec Docs Chain engine docs docs/CONSTITUTION_INTEGRATION.md + docs/USE_CASE_CATALOGUE.md (vasic-digital/docs_chain). Release blocker. No --ad-hoc-sync-ok / --skip-docs-chain / --fake-transform / --sync-evidence-optional escape.

§11.4.107 — Anti-bluff AV/test-validation techniques mandate (User-driven research, 2026-06-02)

Forensic (genericised, 2026-06-02): a test PASSEd on a SINGLE captured frame showing “a picture” — but it was a FROZEN/STALE frame from previously-played content (stale-producer/stuck-decoder), feature broken for the user while green; a sibling FLASHED media on the WRONG output for ~1 s before routing with no test sampling that window; a third shipped a comparator that PASSEd its own deliberately-degraded fixture (the analyzer was the bluff). §11.4.5 mandates captured evidence + a presence pass; §11.4.107 raises it to liveness + correct-routing + self-validated-analyzer.

Every test asserting audio/video output is genuinely playing/advancing MUST satisfy ALL: (1) a single captured frame is NOT proof — prove LIVE, ADVANCING frames over a window via a freeze-detection oracle (near-duplicate/freezedetect-class OR perceptual-hash adjacent-frame distance, NOT byte-identical — byte-identity only a pre-filter); (2) an independent frame-advance counter from the platform’s compositor/decoder telemetry must increase (different-domain oracle — flat ⇒ stuck decoder ⇒ FAIL even if pixels appear to move); (3) loading/buffering is a distinct state — wait for genuine playback-start, never false-FAIL a still-loading stream nor false-PASS a spinner (timeout+unreachable ⇒ SKIP §11.4.3, timeout+reachable ⇒ FAIL); (4) not-stale-from-previous cross-check (new first frame ≠ previous last frame); (5) measured FPS / no-lost-frames within tolerance; (6) no-flash-on-the-wrong-output — high-frequency sampling of the non-target output during routing transitions, any content frame there ⇒ FAIL (content-protection regime classified, never guessed); (7) drive through the realistic feed/UI path, not deep-link shortcuts (UI-not-introspectable ⇒ operator-attended §11.4.52, never fake-PASS); (8) metamorphic relations for the oracle problem on un-owned streams (same content output-A vs output-B must match; paused ⇒ counter stops; 2× speed ⇒ ~2× advance); (9) full-reference quality metrics (SSIM/VMAF/ΔE2000) vs golden source for owned content; (10) mutation-test every analyzer with a golden-good + golden-bad fixture pair so the analyzer cannot bluff (PASS on golden-bad = bluff gate — §1.1 applied to analyzers); (11) per-channel audio RMS/loudness (EBU R128) + XRUN census (aggregate RMS misses a dead channel); (12) OCR overlay/subtitle needs a per-word confidence floor + ROI (avoids both false-FAIL and false-PASS); (13) thresholds calibrated on the project’s own fixtures, not hardcoded from literature (§11.4.6). Honest gap: true photon FPS at the sink has no clean software oracle — flag it, never claim it.

4-layer §11.4.4(b): pre-build gate (CM-AV-LIVENESS-N0-FROZEN-FRAME-class + analyzer-self-validation gate wiring golden-good/golden-bad fixtures into meta-test) + runtime/on-device test + paired §1.1 mutation (single-frame-only → FAILs;

analyzer PASSing its golden-bad fixture → self-validation FAILs) + HelixQA Challenge. Every PASS via `ab_pass_with_evidence` citing motion / not-stale / frame-advance / fps / loudness / metamorphic / self-validation artefacts (`video_display/audio_output/subtitle_render` §11.4.69) — never a single screenshot. Classification: universal (§11.4.17) — platform-neutral, reusable by ANY project validating media playback or any pixel/audio output; project supplies its capture mechanism + calibrated thresholds per §11.4.35. Composes §11.4.5/§11.4.6/§11.4.50/§11.4.68/§11.4.69/§11.4.85 + §11.4.2/§11.4.13/§11.4.48/§11.4.52/§11.4.107. Propagation gate CM-COVENANT-114-107-PROPAGATION (literal 11.4.107) + paired §11.4.107 mutation (gate-code = separate work item).

Canonical authority: Constitution.md §11.4.107. Release blocker. No `--single-frame-proves-playback` / `--skip-liveness` / `--byte-identical-freeze-OK` / `--no-frame-advance-counter` / `--allow-wrong-output-flash` / `--deep-link-shortcut-OK` / `--unvalidated-analyzer-OK` / `--aggregate-rms-suffices` / `--hardcoded-thresholds-OK` escape.

§11.4.108 — Four-layer fix-verification + runtime-signature-as-definition-of-done mandate (systematic-debugging Phase 4.5, 2026-06-03)

Forensic anchor (genericised, 2026-06-03): across one batch multiple fixes were “green” at every gate yet NOT working for the end user — a source-committed change passed by both the source pre-build gate AND the post-build gate never reached the boot-time command-line embedded in the deployed boot artifact (feature dead despite all-green); sibling fixes correctly built into the system image were masked by stale per-user overlay copies from a prior deployment (running code = stale shadow, not the fresh build); deployment did not wipe the mutable overlay so the shadow survived; validation ran against the stale shadow and reported green on code never exercised. Per `superpowers:systematic-debugging` Phase 4.5: each fix revealing a fresh “fixed-but-not-working” in a DIFFERENT place is ONE architectural VERIFICATION flaw, not independent bugs — patching symptoms is thrashing. A fix crosses FOUR distinct layers, and “fixed” at one does NOT imply fixed at the next: (1) SOURCE (committed — `grep-the-source` gate), (2) ARTIFACT (the change’s BYTES landed in the produced artifact — image / bundle / installer / embedded command-line), (3) RUNTIME-ON-CLEAN-TARGET (active on a CLEAN/fresh deployment, the layer the end user experiences, no stale overlay shadowing it), (4) USER-VISIBLE (works for the end user — §11.4.5/§11.4.69 captured-evidence). The mandate (ALL hold): (1) **runtime-signature-as-definition-of-done** — a fix is DONE only when its declared runtime signature verifies on a CLEAN deployment; source-committed ≠ artifact-contains-it ≠ active-on-clean-target ≠ user-visible-working; (2) every fix declares ONE machine-checkable runtime signature (running-system property / sink-side report per §11.4.13 / §11.4.5/§11.4.69 captured-evidence / counter delta — NEVER a re-grep of the source); this registry of per-fix runtime signatures is the SINGLE SOURCE OF TRUTH for “fixed”, REPLACING “a gate greps the source”; (3) gates span all four layers — source (pre-build), artifact (post-build — assert the BYTES landed, not merely that the source still contains the change), runtime-on-clean-target (post-deploy — assert the runtime signature), user-visible (§11.4.5/§11.4.69); (4) eliminate the stale-deployment / shadow layer by construction — deployment MUST yield a CLEAN state (wipe the mutable overlay) OR a pre-validation assertion MUST prove `running-artifact == built-artifact` before any validation; validation against possibly-stale state is INVALID and its PASS is a

§11.4 PASS-bluff; (5) meta-rule — ≥ 3 “fixed-but-not-working” discoveries in one cycle signal an architectural VERIFICATION flaw, not three independent bugs; on the 3rd STOP patching symptoms (§11.4.4), fix the VERIFICATION pipeline, re-certify EVERY item through it on a clean target; (6) a batch is “validated” only after COMPREHENSIVE per-item runtime-signature verification on a clean baseline, NOT after the touched items’ own tests pass. Classification: universal (§11.4.17) — platform-neutral verification-pipeline disciplines reusable by ANY project that builds → deploys → validates an artifact; the project supplies its artifact format, clean-deployment / equal-artifact mechanism, and per-fix runtime-signature observables per §11.4.35. Composes §11.4.1/.2/.4/.5/.6/.27/.40/.46/.50/.52/.69/.102. Propagation gate CM-COVENANT-114-108-PROPAGATION (literal 11.4.108) + recommended per-fix gate CM-RUNTIME-SIGNATURE-REGISTRY + paired §1.1 mutations (gate-code = separate work item).

Canonical authority: Constitution.md §11.4.108. Release blocker. No --source-green-is-done / --skip-artifact-byte-check / --validate-against-running-state / --no-clean-deployment / --skip-runtime-signature / --spot-validate-touched-only escape.

§11.4.109 — Mandatory Anti-Forgetting Enforcement: PreToolUse Guard Hook + Subagent Constitutional Preamble + Orchestrator Pre-Action Checklist (Operator mandate)

UNCONFIRMED forensic anchor (pending operator’s verbatim mandate quote). Background: emulator subagents ran raw host-direct emulator/adb because the orchestrator forgot to inject the Containers-submodule rule at dispatch. A forgotten rule is not enforcement. Two-layer fix: (A) portable bash/awk PreToolUse hook (constitution/scripts/hooks/guard-forbidden-commands.sh, no jq, no project-specific paths) blocks host-direct emulator / force-push bypass / sudo / host-power at the tool-call boundary regardless of agent memory (the floor); (B) constitution/docs/AGENT_GUARDRAILS.md — canonical preamble the orchestrator pastes verbatim into every subagent dispatch + Orchestrator Pre-Action Checklist (the ceiling). Consuming projects wire the hook in .claude/settings.json or equivalent; maintain docs/AGENT_GUARDRAILS.md with SUBAGENT CONSTITUTIONAL PREAMBLE, ORCHESTRATOR PRE-ACTION CHECKLIST, and the anchor literal 11.4.109; provide a ≥ 20 -case hermetic hook test. The hook is inherited by reference — NEVER copied locally. Gates: CM-ANTI-FORGETTING-ENFORCEMENT + CM-COVENANT-114-109-PROPAGATION + paired §1.1 mutations (gate-code = separate work item). Classification: universal (§11.4.17). Composes §11.4.6/.10/.75/.76/.78/.79/.80/.81/.84/.98/.102/§12.

Canonical authority: Constitution.md §11.4.109; ref impl constitution/scripts/hooks/guard-forbidden-commands.sh + constitution/docs/AGENT_GUARDRAILS.md. Release blocker. No --skip-pretooluse-hook / --no-guardrails-doc / --anti-forgetting-optional / --single-layer-sufficient escape.